

MT 204: Thermodynamics and Kinetics of Materials

Course Code	MT 204
Course Name	Thermodynamics and Kinetics of Materials
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	B.Tech.
Mutual Exclusion	None
Approval	57th BoA (Oct 2024)

Course Contents

- **Introduction:** Introduction to thermodynamics: enthalpy, entropy, heat capacity, specific heat, and an atomic view, Systems, states, and material properties; Processes and the First Law of Thermodynamics, Irreversible processes, the Second Law of Thermodynamics, and equilibrium, The combined statement and differential forms, equilibrium conditions, thermodynamic potentials, Helmholtz free energy, Gibbs free energy, chemical potential, thermodynamic relations, and Maxwell's relations and their applications. (10 Hours)
- **Free Energy Functions and their significance:** Clausius-Clapeyron Equation: Derivation of Clausius-Clapeyron Equation and its Integration; Ramsay-Young, Trouton's and Duhring's rules; Fugacity, Activity and Equilibrium Constant, Numerical calculations involving equilibrium constant; Variation of equilibrium constant with temperature, integration of free-energy equation, Sigma function, Concept of Chemical Potential, and Gibbs Phase Rule. (8 Hours)
- **Thermodynamic solutions and Phase diagram:** Thermodynamic stability of materials. Ellingham diagram and its importance, application of electrochemical series. Behavior of solutions: ideal solution, Gibb's-Duhem equation, Raoult's and Henry's law, activity of a component, regular solutions, free energy-composition diagrams for ideal and regular solutions. Phase equilibria & phase diagram: Gibbs phase rule, isobaric phase rule and application to unary, binary, and ternary systems, eutectic and eutectoid, peritectic and peritectoid diagrams, lever rule and its application. Sievert's law: residual gases in steel-properties and functions of slags, slag compositions, structure of molten slags, molecular theory, concept of basicity index, ionic theory; thermodynamics of slag-metal reactions. (12 Hours)
- **Kinetics:** Basic kinetic laws, order of reactions, rate constant, elementary and complex reactions, rate limiting steps and Arrhenius equations, theories of reaction rates-simple collision theory, activated complex theory Heterogeneous reaction; Gas-solid, solid-liquid, liquid-liquid and solid-solid systems. Empirical and Semi-empirical Kinetics, Concept of Johnson-Mehl equation, Thermal analysis. (12 Hours)

Text Books

1. C. H. P. Lupis, **Chemical Thermodynamics of Materials**, Elsevier Science, 1982
2. Gaskell, D.R., **Introduction to Metallurgical Thermodynamics**, 3rd Edition, McGraw-Hill, 1995.
3. Ghosh A., **Textbook of Materials and Metallurgical Thermodynamics**, Prentice Hall of India, 2003.
4. Balluffi R.W., Allen S.M. and Carter W.C., **Kinetics of Materials**, John Wiley & Sons, 2005
5. Upadhyaya G.S. and Dube R.K., **Problems in Metallurgical Thermodynamics and Kinetics**, Pergamon Press, 1985.

References

1. Ahindra Ghosh, **Text Book of Materials and Metallurgical Thermodynamics**, Prentice Hall of India Pvt. Ltd., 2003.
2. Tupkary R.H., **Introduction to Metallurgical Thermodynamics**, TU Publishers, 1995.